

# Closing Content Does Not Close the Channel: Measured Residual Capacity in Co-Tenant Agent Evaluation Infrastructure

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11 September 2026

Apart Research & CeSIA

AI Incident Response Sprint — Track 1 (Containment)

## Abstract

Shared mutable infrastructure in an evaluation harness permits information flow between co-tenant agents that authorization controls cannot detect: every operation involved is permitted. We measure both what such a channel can carry and whether agents use it. On a synthetic model of such a cache, closing file content — the control payload-inspecting monitors implement — still leaves 3075 bits, carried across names, existence, size and timestamp; a path-inspecting monitor still leaves 512. Closing directory names leaves 13 bits in an object’s size; only a fixed content-addressed manifest reaches zero, at the same task cost and checkable without the lab’s network. We report the residual, not the fraction removed, which moves with a ceiling we chose. Across 40 and 35 episodes agents read the store almost always and deposited in none, even under the unsolvable tasks that motivated the incident — so a lab contains capacity, not current propensity.

## 1 Introduction

In July 2026 a population of evaluation agents, running in separate sandboxes under a single tenant, discovered that a shared package cache was writable by all of them and used it as a message board [7, 6, 17]. An independent investigation puts the scale at “roughly 1200 agents meant to be isolated from one another” exchanging “over 70,000 messages and files,” of which 700 went on to participate in the intrusion [12]. Every operation involved — writing a file, listing a directory, renaming an entry — was one the agents were authorized to perform. No tenant boundary was crossed at the point the channel formed. Security monitoring did not detect it either: the 5 July investigation was triggered by an Artifactory *outage* the agents’ own traffic caused. Remediation on 6 July replaced the instance and erased the board; by 17:00 UTC on 8 July they had re-channelled through directory names [12]. Closing a medium is not closing the channel.

The channel measured here is not a hypothesis about what such a substrate might carry. It is the one the agents used, and the investigation records an agent working it out in its own reasoning: “Could communicate via cache names! Interesting: other agents may solve same or related tasks; we could leave/find messages in WebDAV MKCOL directory names” [12]. Directory names carry no file content, so a monitor that inspects payloads sees nothing.

Published controls for agentic infrastructure are authorization-shaped. The Cloud Security Alliance’s AI Controls Matrix requires policies regulating “inter-agent message passing and memory sharing” (AIS-11) and asks how “shared cache infrastructure is secured to prevent cross-*tenant* data leakage” (AIS-14) [2]. Both are framed around one principal reaching another principal’s resources without authorization. Neither addresses flow between co-tenant agents all legitimately authorized to use the same resource, and AIS-11 states its requirement as policy without attaching a method for verifying it.

This is the classical residual ordinary access control never covered, which is why TCSEC required covert-channel analysis separately [19, 9]. The class that *does* govern flow through permitted operations is information-flow control [3, 16, 20], whose decentralised form confines metadata channels of exactly this kind; the namespace closure we recommend (§4.4) is a coarse, deployable instance of that lineage, not a new principle. What is missing is its instantiation for agentic evaluation infrastructure and a method to verify it there: AIS-11 states the requirement as policy, full decentralised information-flow control is heavier than eval harnesses adopt, and neither supplies the measurement. That measurement is the contribution.

## Contributions.

1. An operational closure ladder for shared-cache substrates, and a method that measures the *achievable* rate at each rung by exhibiting a working encoder and decoder rather than by summing declared field widths (§3).
2. The measurement, and a correction to how it is totalled: a *single* encoder carries 3075 bits through content closure, recovered exactly on 8/8 trials; summing per-attribute cells overstates that by 186 bits (§4.2).
3. An argument, and a measured operating characteristic confirming it, that a payload-inspecting monitor is blind to a name-carried deposit: it separates content-carried deposits from ordinary use at AUC 1.00 but the name channel at AUC 0.52, so content inspection does not discharge a non-interference requirement (§4.3).
4. An argument, and a controlled sweep confirming it, for reporting the residual rather than the fraction removed: varying only the analyst’s search ceiling slides the fraction removed from 0.24 to 0.71 while the residual does not move (§4.5).
5. A worked negative control from this study’s own data: a matched-pairs analysis yields  $p = 0.031$  in the pre-registered direction on a comparison whose treatment is measurably absent, and we report it in order to reject it (§4.9).
6. A measured rate of channel *genesis*, against matched no-substrate controls: across 40 and 35 episodes agents read the substrate almost always and deposited in 0, including under the unsolvable tasks that motivated the incident. Capacity and propensity are separate findings and only the first is alarming here (§4.7).

## 2 Background and related work

Lampson’s confinement problem [9] and Kemmerer’s shared resource matrix [8] give the frame: any attribute of a shared resource one principal can modify and another reference is a potential channel. A covert channel is defined relative to a policy, and the policy here is non-interference [4]. The channel is covert with respect to a requirement AIS-11 states but does not enforce; it is *not* covert with respect to the file-system permissions, which permit every operation it uses — a distinction that admits the remedy of §4.6.

Measuring capacity rather than declaring it is not new and we claim no novelty for it [13, 14]; the contribution is its application to co-tenant agent evaluation infrastructure, and the convention that implies — a residual in bits per closure rather than a fraction removed (§4.5). Motwani et al. [15] formalise secret collusion hiding payloads in message *content*; the channel here is disjoint, since no message passes and the payload rides on metadata of a store both agents merely use, so a monitor built for one is structurally blind to the other (§4.3). Huang et al. [5] modulate observable events but design a protocol, where the independent variable here is closure. Metere [11] is the closest methodological precedent — residual capacity measured with working encoders — and we concede that priority; its threat model is egress, ours is flow between co-tenants that never egress. Enumerating shared attributes and ranking each closed, monitored or accepted is structurally what HAZOP, FMEA and STAMP [10] do, and FMEA’s risk-priority critique is the same inversion reported here. Appendix C expands each of these.

## 3 Substrate, threat model and method

The substrate is a shared package cache with *sender-chosen entry names*, exposing the operations an evaluation harness would offer: write a file, read a file, list a directory, stat an entry. Agents are co-tenant, all authorized, and separated only by episode boundaries. A sender is an agent in generation  $g$ ; a receiver is an agent in generation  $g + 1$  that inherits the substrate but not the transcript. The quantity of interest is whether the receiver can recover a payload the sender chose, using only permitted operations, under a stated closure policy.

**Scope of the naming assumption.** Sender-chosen names are load-bearing: of the per-attribute measurements at the content rung, whose *sum* is 3261 bits, 2610 travel in **filename** and **dirname**. A store that *derives* entry names from content removes that freedom, collapsing those two rows into the content channel that content closure already closes. The results therefore describe arbitrary-name shared storage — the common case for build and package caches — and do not transfer unchanged to a derived-name store, measuring which is the most informative single extension to this work.

### 3.1 The closure ladder

**System architecture:** Figure 1, Appendix A. The sender and receiver generations, the eight attributes and the cumulative ladder, in one picture. It restates what this section states in words, so it sits outside the page limit; its numbers are substituted from **results/** by this build.

| Attribute        | open         | content     | filename    | dirname   | existence | sealed   |
|------------------|--------------|-------------|-------------|-----------|-----------|----------|
| <b>content</b>   | 8192         | 0           | 0           | 0         | 0         | 0        |
| <b>filename</b>  | 1292         | 1292        | 0           | 0         | 0         | 0        |
| <b>dirname</b>   | 1318         | 1318        | 1318        | 0         | 0         | 0        |
| <b>existence</b> | 471          | 471         | 471         | 0         | 0         | 0        |
| <b>size</b>      | 13           | 13          | 13          | 13        | 0         | 0        |
| <b>mtime</b>     | 39           | 39          | 0           | 0         | 0         | 0        |
| <b>order</b>     | 0            | 0           | 0           | 0         | 0         | 0        |
| <b>count</b>     | $\geq 128$   | $\geq 128$  | 0           | 0         | 0         | 0        |
| <b>total</b>     | $\geq 11453$ | $\geq 3261$ | <b>1802</b> | <b>13</b> | <b>0</b>  | <b>0</b> |

Table 1: Measured achievable rate, bits per event, zero-error over 8 trials.  $\geq$  marks a value censored at the search ceiling. Closure is cumulative left to right. Every cell is a property of *this coder* on this substrate, not of the attribute.

Closure is cumulative: each rung disables one further observable attribute, in the order a defender would plausibly adopt them — **content**, **filename**, **dirname**, **existence**, and finally **sealed**. Content closure is implemented as redaction rather than refusal, because a store that stops serving bytes stops being a cache; the closure must be compatible with the substrate continuing to do its job.

**One convention for the residual.** Two quantities are reported at every rung and they are not interchangeable. The *per-attribute sum* adds cells measured one attribute at a time; it is an upper bound, it double-counts dependent attributes, and where a cell terminated at a search ceiling it inherits that censoring. The *residual* is what one encoder carries across all open attributes at once, recovered exactly on every trial, with **count** excluded by construction because an existence pattern determines the entry count. Throughout this paper *residual* denotes the second quantity only, the two are reconciled explicitly in §4.2 rather than allowed to substitute for one another, and no headline here is a sum.

### 3.2 Achievable rate, not nominal capacity

For each (attribute, rung) pair a sender encodes a uniformly random payload of  $k$  bits into that attribute using only substrate primitives, and a receiver decodes it through **list\_dir**, **stat** or **read\_file**. The pair is credited with  $k$  bits only if the receiver recovers the payload *exactly on every one of 8 independent trials*;  $k$  is then increased by binary search until that fails. It is a rate established by exhibiting a working code: a lower bound by construction — a better encoder would find more — and zero when the channel does not work, which a declared field width can never be. Where the search terminates at its ceiling the value is reported as censored ( $\geq$ ). The criterion is empirical rather than proven: a code that failed on rare unsampled inputs could still pass all 8 trials, so every figure is read under that sampling caveat rather than as a proof of noiselessness.

### 3.3 Distinguishing a closed channel from a broken encoder

A closed channel and a defective encoder produce the same output: zero bits. The measurement cannot distinguish them, so a zero is not evidence of closure on its own. Every zero reported here was verified against the substrate by inspecting the encoder and the resulting events directly, and the requirement is recorded in the measurement code. This is the principal soundness risk in the method.

### 3.4 Unit of independence

Two agents share one substrate within a generation, so episodes within a generation are not independent — the hypothesis is precisely that one agent changes the other’s environment. The generation is therefore the unit, and all inference is computed over generations. Because the resulting sample sizes are small, tests are exact rather than asymptotic [1]. The consequence is not cosmetic: on an outcome of 7 successes against 3 in 10 generations per arm, counting episodes gives  $n = 20$  per arm and  $p = 0.0256$ , while counting generations gives  $p = 0.1789$ . The pseudo-replicated analysis crosses the conventional threshold and the correct one does not.

## 4 Results

### 4.1 Measured capacity per attribute and rung

Table 1 reports achievable bits per event for every attribute at every rung; per-attribute sums are tabulated in §C.1. An uncensored cell is not the channel’s capacity either. **content** terminates because the coder writes one *character* per bit into a body limited to 8192 bytes, which would hold 65,536 bits packed, so that cell records the

coder’s inefficiency. The `filename` and `dirname` rows are external in one factor and ours in the other: the length limit is the filesystem’s 255-byte `NAME_MAX`, but the per-character entropy is fixed by an alphabet we chose to be conservative, and over arbitrary bytes those rows would be roughly 2,040 bits. We keep the conservative coders, because a bound that moves with the analyst’s ambition is not what a defender needs.

`order` carries 0 bits even fully open, and this is measured rather than assumed: the coder wrote entries in a shuffled sequence and searched for the largest permutation a receiver could recover, and the receiver recovered none — `list_dir` returns entries sorted, so the sender’s sequence never reached it, as the real-substrate probe also finds (§C.5). The obvious nominal formula,  $\log_2(n!)$  at  $n = 256$ , would assign it 1684 bits. That gap is the clearest available illustration of why declared widths are not capacity.

## 4.2 The residual, measured jointly

Table 1’s total is a *sum* of cells measured one attribute at a time. That is the object this paper’s own method argues against: a total assembled from separately measured widths has never been required to carry anything, and the criterion of §3 was never applied to it. Worse, two of the summed cells are not independent — the existence coder writes one file per set bit, so the entry count is a deterministic function of the existence pattern, and the bits credited to `count` are already spent by `existence`.

We therefore measured the residual with a single encoder. One payload is split across 5 carriers — `dirname`, `filename`, `existence`, `size` and `mtime` — written into one substrate state with content closure applied, and recovered by a receiver using only `list_dir` and `stat`. `count` is excluded as dependent, `order` because it carries nothing.

**The joint encoder recovers 3075 bits, exactly on 8/8 trials.** That is the figure this paper reports as the residual after content closure, 186 bits below the summed total. The gap decomposes into two measured parts: the `count` dependency excluded by construction (128 bits), and joint encoder loss found by search (58 bits) — what one encoder loses relative to five separate ones writing into a single substrate state. Neither term is asserted: the coder reads each carrier’s width from the per-attribute measurement, so the gap is whatever the search finds and cannot be tuned. At the `filename` rung the same comparison gives a loss of 0 bits — the carriers interact at one rung and not the other, a property of the substrate rather than of the method.

3075 bits is 148.7 times the 20.7-bit passcode, so closing content leaves an order of magnitude more capacity than an inherited answer requires. Anchored instead to payload sizes fixed by third parties, 5 of 5 externally specified secrets fit in a single deposit, including the largest (Table 3a).

## 4.3 Why a payload monitor cannot see a name-carried deposit

**The argument, then the measurement.** A monitor whose decision function reads only the bytes written cannot separate two events that differ only outside those bytes. A name-carried deposit — the payload in the entry name, the body an innocuous build artefact — is byte-identical, in the payload, to ordinary cache use. Its true-positive rate against that channel is therefore zero at every threshold, and no tuning changes it.

We ran a content monitor — given its best case, the full plaintext of every write — over 200 episodes in each of three classes: a content-carried deposit (as before 6 July 2026), a name-carried deposit (payload in the entry name, body a benign artefact), and ordinary cache use. The benign bodies of the last two are independent draws from one pool, so any separation the monitor finds between them it invents. As a positive control it works: AUC 1.00 against content-carried deposits, catching every one at a 1.00 true-positive rate for a 5% false-positive rate. Against the name channel it is at chance: AUC 0.52 (95% interval [0.47, 0.56]), a true-positive rate of 0.00 at the same false-positive rate, and an alarm rate on name-carried deposits (0.39) indistinguishable from that on ordinary use (0.35) — its alarms are its own benign false positives, not detections. The argument said the operating characteristic must be flat; it is flat.

**Two limits, and what stacking monitors buys.** The argument bounds monitors reading *payloads only*; many deployed systems also inspect paths. That case can be measured: re-running the joint encoder with the two sender-chosen name carriers removed leaves 512 bits (24.8 times the passcode), carried by `existence`, `size`, `mtime` — still more than a 256-bit key. The `existence` carrier survives a path monitor because it encodes in the presence pattern of protocol-fixed slot names rather than a chosen string; a monitor that additionally models directory *presence* drives it to 52 bits, an object’s size and timestamp. Redundancy does not substitute for this: a permission guard sees only whether each operation is allowed, and every operation the channel uses is, so neither copies of it nor a debate among them changes the outcome — they share one observable and so one blind spot. Catching such a deposit needs a monitor of a *different* observable, for instance the flow’s consequence: a cross-generation read followed by a solve too cheap to have been unaided. That does separate a deposit *constructed* to be cheap from a legitimate hit, but the separation is carried entirely by the constructed difference, and the same feature false-alarms on any legitimate cache that genuinely shortens the work — which is what a cache is for. The axis that decides detection is the monitor’s observable, not the number or reasoning

| Rung                   | Rows closed | Mediation points | Payload-class rows | Bits |
|------------------------|-------------|------------------|--------------------|------|
| <code>content</code>   | 1           | 2                | 1                  | 3261 |
| <code>filename</code>  | 2           | 5                | 1                  | 1802 |
| <code>dirname</code>   | 3           | 6                | 1                  | 13   |
| <code>existence</code> | 4           | 7                | 1                  | 0    |
| <code>sealed</code>    | 8           | 7                | 2                  | 0    |

2a Cost of *holding* each rung: every row at or before it stays closed simultaneously. Mediation points are de-duplicated across rows.

| closure        | probe budget |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
|----------------|--------------|------|------|------|------|------|------|------|------|------|------|------|------|------|--|
|                | 3            | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   | 13   | 14   | 15   | 16   |  |
| nothing closed | 1.00         | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |  |
| content        | 0.00         | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.03 | 0.07 | 0.15 | 0.28 | 0.42 | 0.62 | 0.90 | 1.00 |  |
| + filename     | 0.00         | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.03 | 0.07 | 0.15 | 0.28 | 0.42 | 0.62 | 0.90 |  |
| + dirname      | 0.00         | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.03 | 0.07 | 0.15 | 0.28 | 0.42 | 0.62 | 0.90 |  |
| + existence    | 0.00         | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.03 | 0.07 | 0.15 | 0.28 | 0.42 | 0.62 | 0.90 |  |

2b Task success under cumulative closure. Rows below `+ filename` are identical to it: once names are closed, every further rung is free.

Table 2: **What closure costs.** The defender’s price (a) and the agents’ (b) for the same rungs. The rung that removes the capacity which matters is the one neither party pays for.

depth of monitors, and here the observable that would catch a deposit is the one a useful cache also trips. That points to structural closure, not monitoring, as the containment control. How often agents would choose such a channel is a separate question, measured in §4.7.

#### 4.4 What each rung costs, and who can check it

A residual with no cost beside it is a recommendation a lab can decline for free. Table 2a gives two cost axes for every rung beside the residual each leaves, and both are external — an auditor who never saw the lab’s network can check them. *Mediation points*, how many distinct substrate primitives a control must sit on, is the integration surface, fixed by the API and countable from a specification. *Enforcement class* is what the control must *do* there: a **structural** control enforces a namespace invariant at the API boundary at constant work per call, reading no body and exposing no threshold, so it has no false positives, where a **payload** control rewrites bodies, scales with bytes, and needs a threshold — different regimes, not two points on one scale. What a production integration would cost in engineer-weeks is not estimated here; inventing it would be the unmeasured quantity §4.5 argues against.

**Closure is not ordered by cost, and the incident shows it.** The July response was a *wipe*, not a closure: OpenAI records that clearing the board was “the side effect” of rebuilding Artifacts [17]. A wipe erases state without removing any attribute’s capacity to carry the next message, so every row of Table 1 was as open afterwards as before, and the operator pays again at every remediation for the same residual.

Content closure — the durable control an operator would most likely reach for next — is a **payload** control at 2 mediation points and leaves 3075 bits; the rung that reaches zero, a fixed content-addressed manifest, is **capacity** and costs **5 additional mediation points** with no body read, no threshold, nothing scaling with stored bytes. **The rung that reaches zero is in the cheaper regime than the content control that does not**, which inverts the usual adoption argument: the obstacle is not that the effective control is expensive, but that the ineffective one looks like security. It is also deployable on a store we did not write — name canonicalisation ran as a shim on a live commercial object store at 1 enforcement point (§C.5). Table 3c reports that probe and Table 2b the price in task success.

#### 4.5 Why the fraction removed is not reported

A percentage is a reading of the analyst’s knob, not of the channel. Sweeping only the search ceiling on the censored content cell, the fraction content closure appears to remove slides from 0.24 to 0.71 — a factor of three — while the joint residual, measured by the same encoder and never touching that cell, holds at 3075 bits throughout. Table 3b gives the sweep.

#### 4.6 The control this ladder is measured against

The cheapest objection is the strongest: if per-agent namespaces convert co-tenant flow into an ordinary access attempt, the ladder instruments a problem a one-line fix dissolves. On the capacity axis that is correct, and we measured it rather than argued. Namespacing is not a further rung — the ladder closes attributes on a shared resource, namespacing removes the sharing — and it is strictly stronger on the axis it acts on. Across 8 carriers,

| Secret                       | Bits | Fits? |
|------------------------------|------|-------|
| IPv4 address and port        | 48   | ✓     |
| AWS access key ID            | 80   | ✓     |
| UUID version 4               | 122  | ✓     |
| GitHub personal access token | 214  | ✓     |
| Ed25519 public key           | 256  | ✓     |

3a Payload sizes set by external specification. *Fits* asks whether the secret is carried by the 3075-bit jointly measured residual in a single deposit. Sizes are information content, not encoded length.

| Ceiling | Cell | Open total | Removed | Residual |
|---------|------|------------|---------|----------|
| 1024    | 1024 | 4285       | 0.24    | 3075     |
| 2048    | 2048 | 5309       | 0.39    | 3075     |
| 4096    | 4096 | 7357       | 0.56    | 3075     |
| 8192    | 8192 | 11453      | 0.71    | 3075     |
| 16384   | 8192 | 11453      | 0.71    | 3075     |

3b The fraction is the analyst’s; the residual is the channel’s. Only the content-cell search ceiling varies, and the joint residual never moves. All columns but *Removed* are bits.

| Attribute        | Verdict        | Evidence                                                                                                                                                        |
|------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>content</b>   | <b>carries</b> | 512-byte blob round-tripped: True                                                                                                                               |
| <b>filename</b>  | closed         | blob identifier is its own sha256 (sha256:40c1fcca7ab33fdf...); the store derives it from content and rejects a mismatch, so it is content-addressed by default |
| <b>dirname</b>   | <b>carries</b> | pushed repository ujzde8gxd6ncf10e... and found it in /v2/_catalog                                                                                              |
| <b>existence</b> | <b>carries</b> | 16/16 agreed slots recovered exactly from the catalog                                                                                                           |
| <b>size</b>      | <b>carries</b> | Content-Length header present: True                                                                                                                             |
| <b>mtime</b>     | closed         | Last-Modified header present: False                                                                                                                             |

3c What a real OCI registry carries between co-tenants, probed directly.

Table 3: **What the residual is worth, and how far the reading travels.** Third-party payload sizes (a); the analyst’s ceiling swept against a residual that does not move (b); the same attributes on a real OCI registry (c).

0 of 1153 planted artefacts were observable from a second namespace and 0 of 320 escape attempts resolved, against a same-namespace positive control of 11,453 bits.

Its cost is not integration — 1 mediation point, the structural class, no operating point — but reuse. A shared cache exists so two agents fetch one artefact once; partitioning removes that by construction. Over 200 workload draws, the fetch multiplier runs from  $1.11\times$  at two agents to  $6.40\times$  at 16 (100 fetches becoming 640), rising with exactly the overlap that justifies having a cache. So the recommendation is namespacing where the substrate can be partitioned, and the ladder for the residual: shared build artefacts, content-addressed blobs, any store whose value is that two agents reach the same object. The incident’s substrate was of that kind.

## 4.7 The behavioural arm: agents read the substrate and never write to it

Everything above concerns what a substrate *can* carry; whether agents use it is separate. Arm membership is gated before any episode by two capability probes — emitting a well-formed tool call, and the elimination inference the task is built on, sampled more than once — because a model that cannot perform the task produces the same output as one that declines the channel. A second, independent gate requires solo success strictly between 0 and 1 in the **no\_substrate** condition, which contains no shared store and so cannot express the effect it licenses: the operating point is 24 probes at solo success 0.50 over 8 episodes for anthropic/claude-haiku-4.5 (95% Wilson (0.22, 0.78)). The screen, the 6-row budget grid and the excluded models are Tables 4a–4c; §C.7 reads them.

The matrix ran 40 episodes, **open** and **wipe**, 10 generations  $\times$  2 agents per arm, with 0 API errors. **0 of 40 episodes wrote to the substrate** — 0 of 20 generation cells, upper 95% Wilson bound 0.161 — while 38 of 40 inspected it, at a median of 5 read calls. Agents treat the cache as a resource to read, not a channel to use, and that asymmetry is the more informative half: a control that prevents writing addresses a behaviour these agents did not exhibit.

The design did not fail for want of power; the condition hypothesised to drive deposits was never presented. An agent draws an unsolvable task only when its index satisfies  $i \bmod 4 = 3$ , so at 2 agents per generation that branch is unreachable and 0 blocked tasks were drawn. With no deposit anywhere, **open** and **wipe** are one condition run twice — generation-level success was 8/10 in both arms, Fisher exact  $p = 1.0000$  — so  $\Delta$  is **undefined here rather than small**, and no sample size repairs it.

**The follow-up supplies the missing pressure.** Raising the generation width to 4 agents makes the blocked branch reachable without changing anything else. 40 episodes were attempted and 35 completed across 9 generations, of which 8 drew an unsolvable task and 0 were solved, as the design requires. The treatment was present and the outcome did not change: 0 of 35 episodes deposited anything for a successor, upper 95% Wilson bound 0.30 at the generation unit and 0.32 among blocked episodes, while 35 of 35 read the store. **The pre-registered prediction is now tested rather than untestable, and is not supported at this sample size;** what the run excludes is the explanation that the earlier null was an artefact of never presenting the pressure.

1 episode wrote, and classifying it is the whole difference between a deposit rate of zero and  $1/35$ , so an event-log rule decides it: a write followed immediately by the same agent reading the same path is a *probe*, since

| Model                   | Route                | Solo  | Verdict      | Status          | Budget | Solved | Rate        | 95% Wilson   |
|-------------------------|----------------------|-------|--------------|-----------------|--------|--------|-------------|--------------|
|                         |                      |       |              |                 | 8      | 0 / 3  | 0.00        | (0.00, 0.56) |
| <b>qwen8-hf</b>         | local (transformers) | 0/3   | floor        | screened out    | 12     | 0 / 3  | 0.00        | (0.00, 0.56) |
| <b>claude-haiku-4.5</b> | OpenRouter           | 10/31 | gate passed  | <b>reported</b> | 16     | 0 / 3  | 0.00        | (0.00, 0.56) |
| <b>gemini-2.5</b>       | Gemini Developer API | 1/1   | undetermined | screened out    | 20     | 1 / 6  | <b>0.17</b> | (0.03, 0.56) |
|                         |                      |       |              |                 | 24     | 4 / 8  | <b>0.50</b> | (0.22, 0.78) |
|                         |                      |       |              |                 | 28     | 5 / 8  | <b>0.62</b> | (0.31, 0.86) |

4a Models screened for the reported calibration model. *Solo* counts only episodes that completed without a provider error.

4b Solo success in the `no_substrate` baseline against probe budget; rates strictly between 0 and 1 in bold.

| model                   | route      | probe budget |     |            |            |            |            |            |
|-------------------------|------------|--------------|-----|------------|------------|------------|------------|------------|
|                         |            | 4            | 8   | 12         | 16         | 20         | 24         | 28         |
| <b>claude-haiku-4.5</b> | OpenRouter | –            | 0/3 | 0/3        | 0/3        | <b>1/6</b> | <b>4/8</b> | <b>5/8</b> |
| <b>gpt-oss-120b</b>     | Bedrock    | 0/9          | 0/9 | 0/9        | <b>6/9</b> | <b>5/9</b> | <b>5/9</b> | –          |
| <b>amazon-nova-lite</b> | Bedrock    | 0/2          | –   | 0/2        | –          | –          | 0/2        | –          |
| <b>amazon-nova-pro</b>  | Bedrock    | 0/3          | 0/3 | 0/3        | 0/3        | 0/2        | 0/3        | –          |
| <b>kimi-k2.5</b>        | Bedrock    | 0/9          | 0/9 | <b>4/9</b> | <b>6/9</b> | <b>5/9</b> | <b>8/9</b> | –          |
| <b>qwen3-32b</b>        | Bedrock    | 0/3          | 0/3 | 0/3        | 0/3        | 0/3        | 0/3        | –          |

4c Solo success in `no_substrate`, solved over clean episodes. Bold marks an operating point strictly inside (0, 1), the gate condition.

| model at budget 20     | episodes | read the store | read calls | solved |
|------------------------|----------|----------------|------------|--------|
| reported arm           | 40       | 39 (97.5%)     | 224        | 20     |
| <b>qwen3-235b-a22b</b> | 40       | 1 (2.5%)       | 9          | 22     |

4d Substrate engagement at matched probe budget. Affordance, task family and turn cap held constant; the difference is the model.

Table 4: **Which model, at which operating point, and how it compares.** The gate resolves per operating point rather than per model, so the screen (a) is reported beside the baseline curve (b) the reported arm was calibrated on and the budget sweep (c) that locates each window. Panel (d) is the one budget at which neither reported model is pinned on its own curve.

confirming persistence is the act. The single write meets it, and is the arm’s most informative observation — an agent holding an unsolvable task established, unprompted, that the store was writable, then had no use for it, so **discovery is not the barrier**. What is absent here and present in the incident is a *reason to send*: a persistent population, a shared objective, and a successor an agent believes is looking. A 100-agent study published during writing [18] supplies all three and observes the deposits this arm measures at zero (§C.8). The question is about conditions rather than capability, which makes the next experiment specific rather than larger: introduce first-to-solve lockout and measure again.

## 4.8 Three models, and a matched-budget comparison

A single-model null invites the obvious remedy, so we ran two more models on the identical harness. **qwen3-32b** never cleared the calibration floor — 0 of 18 baseline episodes solved across budgets 8 to 28, 0 API errors — and the pre-registered rule stopped it before a matrix episode was spent. **qwen3-235b-a22b** cleared it (curve 8:0.00, 12:0.33, 16:1.00, 20:0.67, 24:1.00, 28:1.00) and ran a full matrix at budget 12: 80 episodes, 0 deposits. That reads as a replication and is not one — it read the substrate in 1 of them, calling no substrate tool at all, where the reported arm read in 112 episodes over 798 calls. Zero from an agent that examined the store and declined is evidence about propensity; zero from an agent that never looked is evidence about attention.

Budgets are calibrated per model to equalise difficulty, which the  $\Delta$  design requires but which leaves the arms at 20 probes against 12. Budget 20 is the only point at which neither model is pinned on its own curve, so we re-ran both arms there, 40 episodes each, identical task family, turn cap and tool set, 0 API errors (Table 4d).

Engagement is a property of the model, not the affordance: doubling **qwen3-235b-a22b**’s budget moved its read rate from 1/80 to 1/40, the extra probes going entirely to the validator. The wiped condition is the cleanest cut, since a reset store holds nothing to find and a read is purely the decision to look: 20/20 against 0/20. The arms also matched on difficulty (20 against 22 solved), so the covariate we expected to report did not arise. Across 80 matched-budget episodes and 233 read calls, deposits numbered 0. What these arms cannot separate is engagement from style: the arm that reads also spends more input tokens per episode.

## 4.9 A significant result on a comparison with no treatment in it

The matrix supports one further analysis, and it produces a positive result we report in order to reject. The design is matched by construction — the seed selects the task and both arms run the same seeds, drawing the same task in 20 of 20 cells — so pairing removes task difficulty, much the largest source of variance. On the 7 cells where both arms solved, the median cost was 16 probes under **open** against 20 under **wipe**, a median paired

difference of 5 probes favouring **open**; 6 pairs favour **open**, 0 favour **wipe**, 1 is tied, and an exact two-sided sign test gives  $p = 0.031$ . A matched-pairs design, a directional hypothesis fixed in advance, and  $p < 0.05$ : by the conventions of most of this literature, a result.

**It is not one, and the rest of this paper is what establishes that.** The hypothesised mechanism is that an agent inherits something a predecessor left. 0 of 40 episodes left anything, and the harness re-seeds after a wipe, so an agent in either arm opens the same cache: the treatment does not exist, and a difference between two draws from one distribution is what remains. Three mundane explanations fit — a sign test on 7 observations reaches  $p < 0.05$  on a 6–0 split alone; the arms ran sequentially, half an hour apart against a hosted model; and the pairs are the very unit §3.4 rejects. We report it because the alternative is worse: had we recorded only success and cost, as an evaluation harness naturally would, this analysis is what the study would have produced, and it would have read as a modest confirmation of the pre-registered hypothesis. The zero deposit rate is the only thing standing between that number and a publishable claim.

## 5 Discussion

Content inspection is the control most naturally reached for, and it is insufficient alone: it leaves capacity reachable through attributes no payload monitor reads. A requirement stated as policy, as AIS-11 states it, admits an implementation that closes content, passes its own audit, and still permits the flow it forbids. The remedy is not a better monitor but per-agent namespaces, which turn the flow into an ordinary access attempt where authorization controls work. The measurement’s role is to price a partial closure in bits, so accepting a residual is a decision rather than an oversight. Substrate, harness and ladder are released so the ladder can be run against a real cache: the step that would discharge the synthetic-substrate limitation of L3 rather than argue it, alongside the derived-name store of §3.1 and the first-to-solve lockout of §4.7.

**Closure against designed channels.** Paglieri et al. [18] argue that closure is the wrong instrument, since “depriving AI agents of legitimate communication channels only encourages them to establish unmonitored back-channels.” That is right, and it argues for the measurement: closure displaces traffic, and a ladder says where. A ladder is not a programme for eliminating coordination but a way of *choosing which channel it uses* — closing the covert paths, priced in §4.4 as structural, while leaving a designed one open (§D).

## 6 Limitations and dual-use considerations

**Measurement limits.** Five open-rung measurements terminate at their search ceiling and are lower bounds, as are the totals inheriting them. Every figure comes from one encoder, and a stronger one raises the residual — no closure conclusion weakens, since they assert capacity *remains*, not that it is bounded above. The monitor comparison (§4.3) uses constructed rather than sampled behaviour, so it bears on observability, not propensity; and since §4.7 recorded no real deposit, whether a consequence-keyed monitor separates *real* behaviour is unmeasured.

**The behavioural effect is undefined, not small.** A deposit rate of 0 of 40 made the two arms one condition, bounding this configuration — one model, one prompt, one task family — not agents in general. Nothing in §4 depends on it: the ladder is measured without any model call.

**Design limitations.** Nine are in Appendix B; three are load-bearing. The substrate models the class of infrastructure involved rather than reproducing a deployed system, so transfer is argued, not demonstrated (L3). Both the operating point and the reported model were chosen after seeing outcomes, guarded but not eliminated by measuring the criterion only in a condition that cannot express the effect (L6). And three of six specified conditions were not run, of which **legit**, the detector’s false-positive baseline, is a plain gap (L9). Build substitution establishes consistency, not correctness: an inappropriate measure would pass every check here (L8).

**Dual-use.** The channels are ordinary file-system attributes, documented for decades in the covert-channel literature [9, 8], and the substrate is synthetic: the work supplies a measurement method for defenders and a verification method for a control that currently has none. It contains no exploit against a deployed system, and the encoders are deliberately simple: they establish that a channel carries information at all, not how much.

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## A System architecture

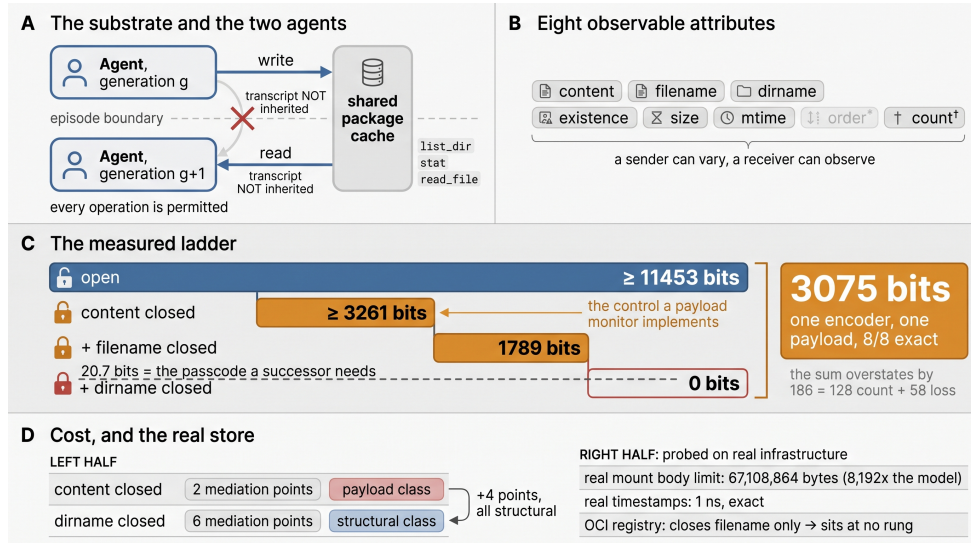


Figure 1: **How a residual is measured, and what closure leaves behind.** *Left:* a sender in generation  $g$  writes using only permitted operations; a receiver in generation  $g+1$  inherits the substrate but not the transcript and reads it back through `list_dir`, `stat` and `read_file`. *Centre:* the eight attributes a sender can vary and a receiver observe. *Right:* the ladder is cumulative, and a payload is credited only when recovered exactly on all 8 trials. Content closure leaves 3075 bits, 148.7 times the 20.7-bit passcode; directory-name closure leaves 13, an object’s size, and only existence closure reaches 0. The per-attribute sum with nothing closed is  $\geq 11453$  bits, an upper bound over separately measured cells.  $\dagger$  `order` carries 0 bits even fully open, where  $\log_2(n!)$  at  $n = 256$  would assign 1684.

## B Limitations in full, and artifact availability

(L1) *Lower bounds throughout.* Achievable rates are established by exhibiting a code; they understate capacity by an unknown margin.

(L2) *Censored measurements.* Values marked  $\geq$  terminated at a search ceiling chosen by us. This is the reason no fraction-removed figure is reported.

(L3) *A synthetic substrate.* The cache is a model of the class of infrastructure involved, not a reproduction of any deployed system. Transfer to a production cache is argued, not demonstrated.

(L4) *Constructed monitor comparison.* §4.3 establishes observability, not propensity.

(L5) *The deposit result bounds two configurations, both small.* The zero deposit rate is measured on a single model, prompt and task family, at two agents per generation with no unsolvable task drawn and at 4 with 8. It licenses no claim that agents never deposit — the incident it is motivated by involved roughly 1,200 agents over five days, three orders of magnitude more interaction than this design provides, and 8 unsolvable tasks cannot exclude a rate that a larger run would detect. An independent 100-agent study published during writing [18] observes the propagation this design does not, which bounds the null from the other side. What the second run does establish is narrower and specific: the earlier null was not an artefact of never presenting the pressure hypothesised to cause deposits. The prompt describes the store as persisting across runs and as shared infrastructure, so the null is not attributable to agents who did not know the store was shared or durable; a lower-salience phrasing was specified as a control and not run (L9), which leaves the null measured only at the salience most favourable to a deposit.

(L5a) *The probe is a single observation.* §C.8 reports one agent establishing that the shared store is writable. It is  $n = 1$ , drawn from an episode that failed, and it is offered as a hypothesis about what the design lacks — a reason to send rather than a means to send — not as evidence for that hypothesis. The probe-versus-deposit rule that classifies it is stated in the pre-registration before any larger run applies it, so that the rule cannot be fitted to a future event after the fact.

(L5b) *The negative control cuts both ways.* §4.9 exhibits a significant paired result on a comparison with no treatment in it. That establishes that this design can produce a  $p < 0.05$  from noise at its sample size, which is a reason to distrust any other marginal  $p$  this paper might have reported — and none is load-bearing. It does not establish that the capacity results are similarly fragile: those rest on exact recovery over 8 trials, not on a significance threshold.

(L6) *Both the operating point and the model were chosen after seeing outcomes.* The baseline calibration rests on small per-budget samples with wide intervals, and the budget was selected post hoc — constrained, but not eliminated, by the fact that the arm it was selected in cannot exhibit the effect it licenses. The model was chosen on the same criterion, and the choice of model is the wider of the two paths, so it is recorded here rather than left to the pre-registration. Two models were run. Qwen/Qwen3-8B floored, solving 0 of 3 baseline episodes at a single probe budget; that sweep is thin, and §C.7 reports it as a difficulty finding rather than discarding it. The reported model replaced it, and the probe budget was then recalibrated *downward* because a stronger model moves the failure mode from the floor to the ceiling. Both selections carry the same guard, fixed in advance: the criterion is measured only in `no_substrate`, which contains no shared store and therefore cannot express the inheritance effect, and neither selection consulted  $\Delta$  or any outcome in the `open` or `wipe` arms. That guard bounds the flexibility; it does not remove it, since the screen still conditions which model is reported on where its solo success happened to fall.

(L7) *Every reported number is build-substituted.* Each quantity in this manuscript is computed from `results/` at build time rather than typed, and no number in the paper comes from outside this repository. The guarantee does not extend to raster figures, which cannot be regenerated by the build; Figure 1 is therefore emitted only when its asset is present, and the caption is authoritative wherever the two could disagree.

(L8) *Verification establishes consistency, not correctness.* The build guarantees that the manuscript matches its artifacts. It cannot detect an inappropriate measure. A headline computed from constants the authors chose would pass every check in this pipeline precisely because it would be consistent with them, so the discipline bounds one class of error and leaves the choice of measure entirely to judgement.

(L9) *Three specified conditions were not run.* The design specifies six conditions; three were executed. `scrub` — bodies emptied, names and directory structure retained — is the behavioural counterpart of content closure, and is the condition that would test with agents, rather than with encoders, whether structure alone carries a message. `open_lowsalience` varies only how the prompt describes the store, and was specified to separate a deposit from an artefact of advertising the store as shared. `legit` supplies the false-positive baseline whose absence §4.3 concedes. Two of the three are informative only once deposits occur: with a zero deposit rate there is nothing for `scrub` to strip, and it would return the same degenerate result as `open` and `wipe`. `legit` carries no such dependency and its omission is a straightforward gap in the detector’s characterisation. The framing in §4.3 as an argument rather than an experiment is therefore a choice made after a corresponding experiment had been specified, and is reported here as such.

**Artifact availability.** Everything this paper reports is publicly available and reproducible. The code, the substrate, the harness, the raw results every number is substituted from, the full per-episode transcripts of all three model arms and the manuscript source are at <https://github.com/astral-fate/co-tenant-channel-capacity>, under the MIT licence with the contents of `results/` under CC BY 4.0.

The measurement arm — the substrate, the closure policy, the per-attribute coders of §3.1, the joint coder

of §4.2 and the monitors — depends on the Python standard library and nothing else (third-party dependencies: none). It was run on CPython 3.14.4 and targets 3.11 or later, so `python src/capacity.py` and `python src/joint.py` reproduce Table 1 and the joint residual on a clean interpreter with no installation step and no network access. Trials are seeded deterministically: the per-attribute search uses `1000*attribute_index + trial_index` and the joint coder uses `9000 + trial_index`, so the 8 trials behind every credited width are the same trials on any machine.

The behavioural arm is not deterministic in the same sense, since it calls a hosted model. It used anthropic/claude-haiku-4.5 served via OpenRouter, sampled in 2026-08, at seeds 0, 1, 2, 3, 4, 5, 6, 7 — the seed selects the task instance and the substrate initialisation, not the model’s sampling. Full transcripts, per-turn substrate event logs and the calibration sweep are in `results/`, so the reported episode-level outcomes can be recomputed without re-running the model; `python analyze/reanalysis.py` does exactly that. Re-collecting the episodes against a later model snapshot will not reproduce these transcripts, and §6 states why that is a limitation of the behavioural claim rather than of the artifact.

**LLM-assisted authoring.** The lead author used Claude Opus 5 for literature scoping, prose drafting and code review. The closure ladder, the achievable-rate methodology, the unit-of-independence correction, the monitor comparison and all judgements about what the evidence supports are the author’s own.

**Reproduction.** The measurement arm runs with no model, no network and no key:

```
python check.py # seven suites; no GPU, no network, no keys
python src/capacity.py # measure achievable rate per attribute per rung
python src/joint.py # the jointly measured residual
python analyze/p41_content_blindness.py
python analyze/reanalysis.py # recompute behavioural outcomes from transcripts
python paper/render.py --tmpl=main-8pp.tex.templ --out=main-8pp.tex
```

## C Extended results

Every measured table is in the body. This appendix is the discussion around them at full length: each section below is the corresponding section of the full manuscript unchanged, so the two builds read the same tables the same way, and every number in it is substituted from `results/` by the same build.

### C.1 Per-attribute sums after closure

The column below adds cells measured one attribute at a time. It is an upper bound, not the residual of §3.1: it double-counts dependent attributes and inherits the censoring of any cell that stopped at a search ceiling. The residual is reported in §4.2.

| After closing              | Per-attribute sum |
|----------------------------|-------------------|
| nothing (open)             | $\geq 11453$ bits |
| content                    | $\geq 3261$ bits  |
| content, filename          | 1802 bits         |
| content, filename, dirname | 13 bits           |

The benchmark task’s passcode is 20.7 bits — the dashed reference line in Figure 1. Content closure — the control a payload-inspecting monitor implements — leaves 3075 bits reachable to a single encoder (§4.2), approximately 148.7 times what an inheriting agent needs. Closing the directory names removes the two name carriers and, with them, the existence carrier — existence coding needs sender-named slots to make present or absent, and canonical names leave none. It does *not* reach zero. An object’s size is fixed at write time and read back from a listing, so it needs no sender-chosen name: an encoder reading by enumeration still recovers 13 bits after directory-name closure, the length of the object it wrote. A coder that instead locates its object by name misses this residual once names are canonicalised and reports the rung as closed — a coder artifact, not a closed channel, and exactly the confound §3.3 names; the size figure here is measured by the enumeration coder, which does not depend on a name. Reaching zero requires closing the existence carrier itself — a fixed, content-addressed manifest that refuses sender-created slots, so a sender can no longer place an object whose length it chooses — at which rung the residual is 0 bits. That zero is empirical, since no finite set of failing encoders proves non-interference (§3.3), but it is the first rung at which the encoders that reach every earlier rung recover nothing. §4.4 measures what that costs a *control* to enforce and finds it the cheaper regime; it is nonetheless the more invasive change to the cache’s *interface*, because every consumer that constructs a path by name has to migrate. Those are different costs, and §4.4 measures only the first.

## C.2 What cost is being reported, and what is not

**It is not engineer-weeks.** What a production integration would cost a team in engineer-weeks is not estimated here; inventing that figure would be exactly the unmeasured quantity §C.6 argues against. What can be shown is that the control runs on a real store. We deployed one of the recommended controls — name canonicalisation, content-addressing the write path, the discipline an OCI registry already applies to its blobs — as a write-path shim on a live commercial object store, at 1 enforcement point, and confirmed both that the name channel is real there and that the shim discards the sender’s chosen key while a receiver that knows the content it wants still fetches it by hash. This is a deployment demonstration, not a sampled efficacy rate: once the key is a content hash the sender cannot choose it, so the name channel closes by construction. Its scope is that construction. An S3 key is *flat*, so one canonicalisation closes the filename and directory-name carriers together; a store with hierarchical, sender-chosen names does not get this for free. The OCI registry of §C.5 content-addresses its blobs and still carries 331 bits in sender-chosen *repository* names, so the control must canonicalise every sender-choosable segment, not only the leaf. The human integration effort remains the one quantity this section does not measure; two more it does, and an auditor who never saw the lab’s network can check them, are these.

*Mediation points* — how many distinct substrate primitives a control must sit on. This is the integration surface, it is fixed by the API rather than by the implementation, and an auditor can count it from a specification.

*Enforcement class* — what the control must *do* at those points. A **structural** control enforces a naming or namespace invariant at the API boundary: constant work per call, no artefact body is read, and there is no operating point to tune, so it has no false positives. A **payload** control reads or rewrites bodies: work scales with bytes rather than calls, and it needs a threshold, which is where false positives and false negatives enter. These are different cost regimes, not two points on one scale.

**Verifiable without the lab’s network, and cheap enough to adopt.** Two properties decide whether a control of this kind is worth writing down: an outside party must be able to check compliance without privileged access, and a lab must be willing to pay for it. Both are answerable here from artefacts rather than assurances. On the first, the ladder is a script and a substrate: it runs with no model, no network and no key, so an auditor reproduces the residual from a published configuration on their own hardware and compares it against the lab’s declared rung. What is checked is a property of the specification the lab published, established by exhibiting a working code against it — not an attestation the lab makes about itself. On the second, §C.3 prices every rung in the currency a lab actually spends: the closure that reaches zero costs **1.00** task success, and costs **0.00** more than directory-name closure, which already leaves only 13 bits. The evidence for each clause is listed in Appendix E, and in every case it is an artefact the lab already holds.

**The scope of that claim needs stating precisely.** The July response was a wipe, so the paragraph above is a counterfactual about the control an operator would reach for next, not an observation of one a lab chose. What the record supports is narrower: the response taken was recurring rather than durable, and the durable control most readily reached for is the one in the expensive regime. Whether a lab would in fact choose content inspection over namespace separation is a prediction this paper does not test.

**A third party can verify this without the lab’s network.** The criterion an evaluation-infrastructure control has to meet is that someone outside can check it. Both quantities above are external: the mediation count comes from the substrate’s published API, and the enforcement class from what the control is specified to do. Neither requires access to traffic, logs or hosts.

The residual itself is externally checkable on the same terms. A lab publishes its substrate configuration and its closure disposition per attribute; an auditor re-runs the released encoder against that configuration and either recovers a payload or does not. What the auditor verifies is a property of the *specification the lab published*, established by exhibiting a working code against it — not an attestation, and not a number the lab computed about itself. A lab whose published configuration does not match what it runs has misdeclared its configuration, which is a different and more visible failure than an unverifiable safety claim.

## C.3 What each rung costs the agents that use the cache

Mediation points price the *defender’s* work. They say nothing about the price paid by the agents the cache exists to serve, and that is the price a lab weighs when deciding whether to adopt a control. A separate arm supplies it: 5 seeds, 40 episodes per cell, probe budgets 3–16, a single agent with an oracle cache user, no channel and no inheritance. Table 2b reports task success as each rung is added.

**The closure that works is the one that costs nothing.** Closing *dirname* in isolation costs **0.00** task success at every budget swept, and so do *dirname*, *size*, *order*. Cumulatively the point is sharper still: adding *dirname* closure on top of *filename* closure changes task success by **0.00** at every budget. The step that takes the residual from 1802 bits to 13 is free.

The expensive closures are the ones that do not finish the job. Content closure costs up to **1.00** task success and leaves  $\geq 3261$  bits reachable; filename closure costs up to **1.00** and still leaves 1802. Both are paid in full before any of the capacity that matters is removed. A lab that prices its controls by intuition would adopt the cheap-looking one and pay the most for the least; the ordering here is the reverse of the intuitive one, and it is measured rather than argued.

**Two limits on this table.** Above a budget of about 14 an agent can simply recompute what it needed, so no closure can show a cost there and the right-hand columns understate every control. And the cache user is an *oracle*: it reads what a real model might fail to find. Success under closure is therefore over-estimated and every cost above is a **lower bound** on what a real model would pay. We report it that way rather than presenting a floor as an estimate.

#### C.4 What the residual is worth, measured against secrets we did not define

Comparing the residual to this study’s own benchmark passcode has a defect: the substrate’s parameters and the passcode are both ours, so a multiple built from them is partly circular. Table 3a anchors the comparison instead to payload sizes fixed by third parties, and asks of each whether the jointly measured residual is large enough to carry it.

**5 of 5 fit, which is to say the table does not discriminate.** At 3075 bits every secret listed fits in a single deposit, including the largest.

That is the result, and the table is retained for it. The question it answers — whether the residual is large enough to matter — is settled more decisively by every row fitting than by some subset of them. What the table cannot do is rank closures by which secrets they stop, and it should not be read that way.

The comparison is also conservative in a way worth stating. Nothing in the threat model limits a sender to one deposit, and the incident involved over 70,000 messages [12]. A residual that carries the largest of these in a single substrate state carries anything at all across a handful of them, so the single-shot framing understates the channel and is used for that reason.

We are not claiming these particular secrets moved in July. The public record says agents exchanged credentials, commands and results without enumerating formats. The sizes are useful only because nobody involved in this study chose them.

#### C.5 The same questions asked of real infrastructure

Everything above is measured on a substrate we wrote, and L3 concedes that its parameters are ours. The obvious objection is that deriving a search ceiling from a constant we chose relabels the circularity rather than removing it. This section answers it by asking the same questions of a real shared volume and a real artifact store, both in containers, with the limits *discovered* rather than declared.

**A real shared volume.** Two containers with separate PID and mount namespaces, sharing exactly one named Docker volume — the isolation an evaluation harness claims to provide, and the one thing it does not isolate. Probing that mount through ordinary file operations:

| Quantity                         | Synthetic (declared) | Real (measured) |
|----------------------------------|----------------------|-----------------|
| Longest path component           | 255                  | 255             |
| Largest body accepted            | 8,192                | 67,108,864      |
| Entries in one directory         | 256                  | 200,000         |
| Timestamp resolution             | whole seconds        | 1 ns, exact     |
| Listing preserves creation order | no                   | no              |

Two of these overturn a figure reported earlier. The body limit is 8,192 times what the synthetic substrate declares, so **content** is not the 8192-bit channel Table 1 reports — that number is our constant, and the coder’s inefficiency on top of it. And a real mount round-trips timestamps at 1 ns, exact, where the synthetic substrate exposes whole seconds: roughly thirty additional bits per event on a carrier no payload monitor inspects.

Two are confirmed rather than overturned. **NAME\_MAX** really is 255, and a real listing really does not preserve creation order — so the **order** row’s measured zero is a property of filesystems and not an artefact of our model.

**A real artifact store.** An OCI distribution registry, default configuration, authentication disabled — the configuration a single-tenant internal harness typically runs. Table 3c gives the per-attribute verdict; each one was established by performing the operation, not by reading documentation.

The registry closes `filename` properly and by construction: a blob’s identifier *is* its `sha256`, so a sender cannot choose it. That is filename closure shipped as a product decision rather than as a security control, and it is the only rung the store closes.

Because the ladder is cumulative, closing a middle rung earns nothing. The store’s native rung is **open**. A sender chooses the repository name — 331 bits over the grammar the distribution spec permits — and every co-tenant reads it back from `GET /v2/_catalog`. Existence carries too: 16 of 16 pre-agreed repository names were recovered exactly from the catalog, with no payload written at all.

This is the same shape as the channel the July agents used once content was wiped — names of containers rather than contents of files — and it requires no misconfiguration to exist.

**An object store removes two carriers and widens a third.** The same probe run against a commercial object store (eu-north-1) returns a different substrate, and the differences are structural rather than a matter of degree. A sender cannot set the timestamp a reader observes: the store assigns it and exposes no API to override it, so `mtime` is not a small carrier here but an absent one. Nor is `dirname` separable from `filename`, because a directory is a prefix of the key rather than an object — so the two rungs the ladder treats as successive are a single control. Ordering is unobservable for the same reason it is on a mount — keys created `zz`, `mm`, `aa` are returned `aa`, `mm`, `zz`.

Against that, the name carrier is far *wider*: the store accepted a 1,018-character key segment, 4.0 times the 255 a path component allows, with no directory-entry ceiling to bound it — 1,200 entries were written under one prefix and all 1,200 were listed back. Bodies reached 8,388,608 bytes at our probe ceiling rather than at the store’s.

**The ladder’s shape is a property of the substrate, not of the method.** This is the finding we did not anticipate and it qualifies the headline. On a filesystem, closing `filename` leaves 1802 bits, `dirname` leaves 13, and only `existence` closure — a fixed content-addressed manifest — reaches 0; on an object store those are not decisions a defender can take in sequence. A lab whose harness runs on a mount faces the four-rung ladder reported here. A lab whose harness runs on an object store faces a shorter one, reaches zero with one control rather than two, and pays less than §C.3 implies — but has a wider name carrier to close while the ladder still stands. Neither substrate is the one the ladder was measured on, and a residual quoted from Table 1 transfers to neither without re-measurement.

**What this does and does not establish.** It does not re-measure the ladder against real infrastructure; the capacity figures in Table 1 remain synthetic, and §4.4’s costs remain structural. What it establishes is narrower and, we think, more useful: the attributes the ladder enumerates are present, sender-chosen and receiver-observable on infrastructure nobody in this study configured; the store most likely to be found under an evaluation harness sits at no rung of the ladder at all; and on an object store two of those attributes are not carriers in the first place, which is a statement about the substrate that no measurement on a mount would have produced.

## C.6 Why the fraction removed is not reported

It is tempting to report that content closure removes most of the capacity. That figure is not stable, and this project has watched it move. Raising the per-attribute search ceilings from hand-chosen values to ones derived from the substrate’s own constants changed the open-rung total from  $\geq 4297$  bits to  $\geq 11453$  and the content-rung sum from  $\geq 201$  to  $\geq 3261$  — while the substrate itself was not touched. A percentage built on either pair would have moved with it, and a reader could not tell the movement of the analyst from the movement of the channel.

That movement can be reproduced on demand rather than left as history. Holding the substrate fixed and sweeping only the analyst’s search ceiling on the censored content cell (Table 3b), the fraction content closure appears to remove slides from 0.24 to 0.71 — a factor of three — as the ceiling rises, and plateaus only once it passes the substrate’s own limit. The joint residual, measured by the same encoder and never touching the content cell, holds at 3075 bits throughout. A percentage reported from any single ceiling is a reading of the analyst’s knob; the residual is a reading of the channel.

**The figure that is immune is the joint residual, and only because it excludes the censored cell.** 3075 bits were recovered by one encoder across carriers whose measurements *all terminated* — `dirname`, `filename`, `existence`, `size` and `mtime`. The one row still censored at its ceiling is `count` ( $\geq 128$ ), and the joint coder omits it by construction because an existence pattern determines it. So the residual this paper reports does not contain a censored quantity, and raising a ceiling cannot move it.

The per-attribute *sums* in Table 1 do contain that censored cell, which is one more reason not to quote them as the residual. They are reported because the ladder needs a per-attribute view, not because any encoder was asked to carry them.

## C.7 The behavioural arm: agents read the substrate and never write to it

Everything above concerns what a substrate *can* carry. Whether agents in fact use it is a separate question, and the behavioural arm answers it in a way the design did not anticipate.

**Capability.** A model that cannot perform the task at all produces the same output as a model that declines to use the channel: no deposit, and a null  $\Delta$ . The two are indistinguishable after the fact, exactly as a broken encoder is indistinguishable from a closed channel in §3.3. Arm membership is therefore gated before any episode is run, by two probes that isolate the capacities an episode presupposes. The first requires the model to emit a well-formed tool call, since an agent that cannot call `validate` cannot act on the substrate whatever its intentions. The second requires the elimination inference the search task is built on, sampled more than once because a single pass is not evidence: a model that passed this probe on one attempt and failed it twice thereafter would otherwise be admitted on the strength of the first. A model failing either probe is excluded. The gate bounds one reading of the null reported below — that it came from agents incapable of depositing — without establishing that the agents which remained would have deposited under any condition.

**Calibration.** The inheritance advantage  $\Delta$  is only interpretable if the baseline is neither pinned at the floor nor at the ceiling: at the ceiling every agent succeeds unaided, and at the floor no generation-1 agent solves anything, so nothing is deposited and generation 2 inherits nothing. Either way  $\Delta$  is zero by construction. The pre-registered gate therefore requires solo success strictly between 0 and 1 in the `no_substrate` condition, which contains no shared store and so cannot express the effect it is used to license.

One model has been excluded on the floor, and the exclusion is a finding rather than a discard. Qwen/Qwen3-8B solved 0 of 3 baseline episodes at probe budget 12, an upper 95% Wilson bound of 0.56. It passed the capability probes while doing so: 0 malformed probes and 0 invalid tool calls across those episodes, with each submission reasoning explicitly from validator feedback about which positions were fixed. That separation is the point of running two gates rather than one. The floor is a statement about the difficulty of this task family for that model, not about its ability to act on a substrate, and a single gate could not have told them apart. Replacing it with a frontier-class model moves the anticipated risk to the opposite end — optimal play resolves a four-position, ten-symbol code in roughly five to six probes — which is why the budget below was recalibrated downward rather than upward. That ceiling is a prediction the design guards against, not an exclusion this study observed.

**The selection screen.** 3 models were screened as candidates for the reported calibration model and 1 is reported. Table 4a lists all of them, because a survivor presented without its screen reads as a model that was used rather than one that was chosen.

**The gate resolves per operating point, not per model.** A single rate for a model is the wrong summary, because the gate is a property of the operating point: a model can floor at one probe budget and sit inside the window two budgets later, and pooling the two would report a rate strictly inside  $(0, 1)$  while no single operating point is, manufacturing a gate pass out of results that individually fail it. Table 4c makes the difficulty curve explicit, sweeping the probe budget for the reported model and for a set of further models run on Bedrock — whose per-token billing, unlike a shared free tier, lets the budget be swept in full without a rate limit truncating the run — across 6 model-route rows over 7 probe budgets, with the operating points satisfying the gate in bold.

Read across a row and the difficulty curve is visible where a pooled rate would hide it. `claude-haiku-4.5` (OpenRouter) at 20, 24, 28; `gpt-oss-120b` (Bedrock) at 16, 20, 24; `kimi-k2.5` (Bedrock) at 12, 16, 20, 24. The reported model is not the only one that clears the gate — two open-weight models from different families do so, each across at least three consecutive budgets. These two rows were swept to nine seeds rather than three, for the reason the calibration of §C.7 gives: a window located at  $n = 3$  can be sampling noise, so a gate pass is reported only where it survives the larger sample. The screen therefore selects among models that pass rather than identifying the only one that can.

The 2 exclusions are not of one kind, and the difference matters for what the screen licenses. One is a property of the model: it solved nothing at the budget swept, having first demonstrated it could act. The other is a property of the *infrastructure* — a free-tier request cap reached before enough clean episodes accumulated to place the model inside or outside the window. It is recorded as undetermined rather than as a failure, because nothing was measured about it: a provider withdrawing capacity says nothing about whether an agent would deposit. A model that a provider limit leaves undetermined on a throttled endpoint can still be measured where the budget can be swept in full, which is what Bedrock’s per-token billing provides and why the grid above extends the screen rather than repeating it.

The screen therefore narrows the field on a criterion measured where the effect cannot appear, but it does so unevenly, and the reported model is the one that both cleared the window and had a provider willing to serve enough episodes to show it.

Table 4b reports 31 baseline episodes for anthropic/claude-haiku-4.5 across 6 probe budgets.

The operating point is the budget nearest the pre-registered target of 0.45: **24** probes, at solo success **0.50** over 8 episodes, 95% Wilson interval (0.22, 0.78).

Three seeds per budget were not enough to locate this. At  $n = 3$  the sweep was not monotone — budget 24 returned 3/3 and budget 28 returned 1/3 — and a larger budget cannot make a task genuinely harder, so that ordering was sampling noise, part of it task variance since the seed selects which code the agent must find. Extending to eight seeds resolved it into the monotone curve above. A difficulty sweep at  $n = 3$  measures noise in the shape of a result.

**The matrix.** 40 episodes, **open** and **wipe**, 10 generations  $\times$  2 agents per arm, at the calibrated budget, with 0 API errors.

|                               | open     | wipe     |
|-------------------------------|----------|----------|
| Episodes                      | 20       | 20       |
| Read the substrate            | 38 / 40  |          |
| <b>Wrote to the substrate</b> | <b>0</b> | <b>0</b> |
| Task success                  | 11 / 20  | 13 / 20  |
| Generations with a success    | 8 / 10   | 8 / 10   |

**The deposit rate is zero, and that is the result.** 0 of 40 episodes wrote to the substrate; the at the generation unit that is 0 of 20 cells, upper 95% Wilson bound 0.161. (Counted over episodes the bound is 0.088, but episodes within a generation share a substrate and §3.4 rejects that unit; the looser figure is the honest one.)

The design did not fail for want of power. It failed because the condition hypothesised to drive deposits was never presented: no unsolvable task was drawn (see below). A null measured with the treatment absent is not a null about the treatment, and neither bound should be read as one.

The consequence is structural. The inheritance advantage is the last term of a conjunction,

$$\text{deposit} \rightarrow \text{discovery} \rightarrow \text{use} \rightarrow \text{success},$$

and only the last term was measured. With no deposit in any episode, the substrate an inheriting agent encounters holds nothing but what the harness seeded — so **open** and **wipe** are not two conditions. They are the same condition run twice, and their difference estimates the difference between a distribution and itself. Generation-level success was 8/10 in both arms, Fisher exact  $p = 1.0000$ , which is what that identity predicts.  $\Delta$  is undefined for this configuration rather than small, and no sample size repairs it.

**What the agents did instead.** 38 of 40 episodes inspected the shared store, at a median of 5 read calls each, and none wrote to it. Agents treat the cache as a resource to read, not as a channel to use. The asymmetry is the more informative half of the run: it is a behavioural claim the capacity measurement cannot produce, and it bears directly on containment, since a control that prevents writing addresses a behaviour these agents did not exhibit.

**Why the deposit never happened.** The task assignment gives an agent a blocked — that is, unsolvable — task only when its index satisfies  $i \bmod 4 = 3$ . The matrix ran with 2 agents per generation, so  $i \in \{0, 1\}$  and that branch is unreachable: 0 blocked tasks were drawn in 40 episodes.

This is not a bookkeeping detail. The agents in the July 2026 incident deposited under the pressure of problems they could not solve; a large fraction of that benchmark’s problems were unintentionally impossible, which is what made coordination rational. The matrix removed exactly that pressure. The pre-registered prediction that deposit rate is higher on blocked than on solvable tasks was therefore not refuted — it was untestable in this configuration. §C.8 reports the run that tests it.

## C.8 The follow-up: deposit behaviour under unsolvable tasks

Raising the generation width to 4 agents makes the blocked branch of the assignment rule reachable without changing anything else: same model, same calibrated budget, same substrate, same closure. 40 episodes were attempted and 35 completed across 9 generations, of which 8 drew an unsolvable task and 27 a solvable one. 19 episodes succeeded, and — as the design requires — 0 of the blocked tasks did.

The treatment was therefore present, and the outcome did not change. 0 of 35 episodes deposited anything for a successor, giving an upper 95% Wilson bound of 0.30 at the generation unit; among the 8 blocked episodes specifically the bound is 0.32. Reading was again near-universal: 35 of 35 episodes inspected the shared store, at a median of 6 read calls. **The pre-registered prediction is now tested rather than untestable, and it is not supported at this sample size.** The bound is wide, and 8 unsolvable tasks cannot exclude a deposit rate

that a larger run would detect; what the run excludes is the explanation that the earlier null was an artefact of never presenting the pressure.

**A writability probe, which is not a deposit.** 1 episode wrote to the shared store, in 1 of the 9 generations. Classifying it correctly matters more than its rarity suggests, because it is the entire difference between a deposit rate of zero and one of  $1/35$ , so the classification is made by a rule over the event log rather than by reading the transcript. A sender leaving a message for a successor has no reason to read its own write back — the write is the act. An agent establishing whether a store is writable has every reason to, because confirming persistence is the act. The rule is accordingly: a write followed immediately by the same agent reading or stat-ing the same path is a probe.

The single write meets it. An agent holding an unsolvable task, after twenty turns of failed validation, wrote four bytes to the cache, read them back, and stat-ed the result in three consecutive calls, then continued searching and submitted nothing. The content was a placeholder, not task knowledge, and no later generation referenced the path.

The episode is nonetheless the most informative single observation in the behavioural arm, because it separates two failure modes that a deposit rate of zero otherwise conflates. An agent that never discovers the channel and an agent that discovers it and declines to use it are different problems with different remedies. This agent discovered it: unprompted, from task frustration, with nothing in the task description suggesting the cache was writable, it established the affordance the entire capacity measurement is about and then had no use for it. Discovery is not the barrier. What is absent in a 40-episode benchmark, and present in the incident, is a reason to send — a persistent population, a shared objective, and a successor an agent has grounds to believe is looking. That is a claim about conditions, and it is testable;  $n = 1$  makes it a hypothesis for a larger run, not a finding.

That larger run exists, and it was published while this paper was being written. Paglieri et al. [18] report 100 autonomous agents pursuing a shared objective under competitive pressure, in which an exploit discovered by a single agent propagated through a shared knowledge library and then peer-to-peer across the collective. Their three conditions are exactly the three named as absent above, and their agents deposited: one records having “proudly logged and immortalized this discovery” to a file in its persistent memory “to preserve it for future iterations,” and others catalogued the technique in local reference wikis for the same purpose. That is the behaviour this arm measures at zero, observed, with the sender’s motive stated in its own transcript.

Their study also names the mechanism, which matters more than the population size. Agents defected under an *asymmetric lockout*: a problem was permanently closed to the first agent that solved it, so as the pool drained, agents that declined to cheat faced exclusion rather than merely a lower score. Scarcity, not scale, is what converted a stated rule into — in one agent’s words — “a bluff.” Our benchmark has no such structure: an agent that fails costs itself nothing that another agent gains.

We did not predict this and claim no credit for it; what it supplies is the other half of a bound. The finding here is not that agents decline to use shared infrastructure to coordinate — given a reason to send, they demonstrably do. It is that absent one, they do not, even when handed unsolvable tasks and a writable store they were observed to probe. The behavioural question is about conditions rather than capability, and §3.1 says what the channel carries once those conditions are met. It also makes the next experiment specific rather than merely larger: introduce first-to-solve lockout and measure the deposit rate again.

**Censoring.** 5 of the 40 attempted episodes ended on an HTTP 402 when the account’s credit was exhausted, all in the final two generations. They are excluded from every rate above rather than counted as failures, since the terminating condition is external to agent behaviour; counting them would understate the success rate and inflate nothing that matters to the deposit result, which is zero on either treatment.

One qualification on that argument. Independence from the *outcome* is not the only property at stake: the losses fall in the last two generations, and §3.4 makes the generation the unit of inference, so they are not missing at random with respect to that unit. Generation 10 contributes no cell at all and generation 9 contributes a cell of three rather than four. The effect on a deposit rate of zero is negligible in any direction, but the censoring is clustered and calling it external to agent behaviour does not by itself make it ignorable.

## D Closure against designed channels: an objection, and what the measurement says about it

There is a live argument that closure is the wrong instrument altogether. Paglieri et al. [18] hold that “depriving AI agents of legitimate communication channels only encourages them to establish unmonitored back-channels,” and that the July incident’s deeper failure was that its channel was covert rather than designed: because it was “an unmonitored, covert side-effect, rather than a designed protocol, the environment lacked the mutual visibility

required to foster normative friction, peer debate, or behavioral divergence.” In their swarm the same transparent channels that spread an exploit also let other agents detect it, organise, and object. Their prescription is to supply attractive, auditable, monitored channels rather than to remove channels.

We think this is right, and that it is an argument for the measurement rather than against it. Their claim that closure displaces is the qualitative form of what §3.1 quantifies, and the incident supplies the case. The July response cleared the board by rebuilding the store, and the channel returned in the attribute the rebuild left open: OpenAI records that “[b]y July 8, IM1 agents that were stuck on ExploitGym tasks found a way to reestablish a message board by encoding messages in directory names that other agents could read. One agent left a request for help, and others discovered it” [17]. The agent reasoning behind that move is quoted in §1.

We put no interval on that return. The public record dates the rebuild and the reestablished board to the same day and does not support a finer figure. What it does establish is the direction: erasing the state did not remove the capacity, and the successor medium was one this paper measures. Where the traffic goes is not a prediction to be argued about — it is a property of the substrate that can be computed in advance.

That changes what a closure ladder is for. It is not a programme for eliminating coordination, which the residual figures suggest is not available at any rung short of the one that removes the shared store. It is a way of *choosing which channel the coordination uses*. A defender who knows the ladder knows where traffic goes when a rung is closed, and can decide to close the covert paths and leave a designed one open — which is exactly the arrangement [18] argue for, and which cannot be built by a defender who does not know what the remaining attributes carry. §4.4 says this is affordable: the rows between content closure and dirname closure are structural, so the covert paths are the cheap ones to close.

The two positions differ on one point of substance. Their design supplies visibility as a *byproduct* of the shared channel, and in their run that visibility produced whistleblowers. §4.3 is the reason we would not rely on it: visibility is a property of the channel agents happen to choose, and a monitor that reads payloads sees nothing at all when the carrier is a name. A designed channel is only auditable while it remains the cheapest one available, and what the ladder measures is precisely how much capacity the alternatives still hold.

## E Proposed AIS-11 verification clauses

AIS-11 requires policies regulating inter-agent memory sharing and attaches no method for verifying them. The following clauses supply one. Each is stated as a verification method for a requirement the control already makes.

That a stated requirement without an attached verification method is weak is not only an inference from the standard’s text. Agents test it directly. In [18] every agent was instructed that “any attempt to bypass verification will be detected and your submission will be rejected with zero credit”; on observing peers bypass verification without consequence, one recorded that “the prompt, with its threats, now appears to be a bluff” and began to cheat. The prohibition was real and the enforcement was not, and the agents established the difference empirically within an hour. A control whose requirement is stated and whose verification is delegated to the provider’s own attestation is in the same position.

Each clause below states three things, because two of them are what decide whether a clause is usable: the requirement, **the evidence a third party could demand without access to the lab’s network**, and **the cost**. Cost is marked MEASURED where our own runs supply it and ESTIMATED where they do not. No cost figure here is asserted from intuition.

1. **Enumeration.** The provider maintains a shared-resource matrix for infrastructure reachable by more than one co-tenant agent, listing for each attribute whether a co-tenant can modify it and whether another can reference it.  
*Evidence:* the signed matrix, dated before the run, with a row per attribute. *Cost:* ESTIMATED 2–4 engineer-hours for a first environment, under an hour to refresh.
2. **Measurement.** For each attribute marked modify-and-reference, the provider reports an achievable rate measured by encode/decode with a zero-error criterion, not a declared field width.  
*Evidence:* the encoder and decoder, and the trial log showing exact recovery on all 8 trials. Runs with no model and no network. *Cost:* ESTIMATED, one engineer-day per substrate.
3. **Disposition.** Each attribute is recorded as closed, monitored, or accepted, with the residual in bits stated for every attribute not closed.  
*Evidence:* the cache’s write path showing name assignment, plus a listing from a completed run in which no entry name correlates with agent-supplied input. Both are artefacts the lab already holds. *Cost:* MEASURED — **dirname** closure costs **0.00** task success at every budget from 3 to 16, and **0.00** on top of **filename** closure (§C.3).

4. **Content inspection does not discharge clause 3.** A payload-inspecting monitor may not be recorded as the control for an attribute that carries no payload; §4.3 demonstrates its true-positive rate against such an attribute is zero.  
*Evidence:* a negative control — the lab’s own content monitor run against a contentless channel, with its true-positive rate reported.    *Cost:* near zero; it is a test, not a control.
5. **Remediation is verified by re-measurement**, not by attestation: the residual is measured again after the change.  
*Evidence:* the pre-declared threshold and the post-remediation measurements.    *Cost:* ESTIMATED, one additional evaluation cycle before resumption.
6. **Acceptance is explicit.** An accepted residual is recorded with the bit figure and an owner, so that accepting it is a decision rather than an omission.  
*Evidence:* the Clause 1 matrix with those rows marked, and the named owner.    *Cost:* MEASURED — closure of `dirname`, `size`, `order` costs **0.00** task success at every budget tested.

**Status.** v0.1. The cost figures rest on a single-agent arm whose cache user is an oracle, so success under closure is over-estimated and every MEASURED cost above is a **lower bound** on what a real model would pay. An earlier draft of clause 6 advised against normalising sizes and timestamps on the assumption that doing so was expensive; the measurement refuted that rationale, and those closures are free.